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Dialogue systems for language learning: conceptual framework and research challenges

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PhD Day ▪ Leuven ▪ Sept. 11, 2015

Dialogue systems for language learning

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Dialogue-based CALL

- Definition
- Typology

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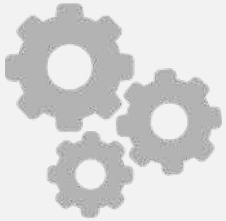
Dialogue systems

- NLP architecture
- Challenges



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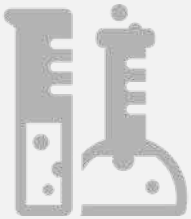
Impact on language learning

- SLA theories & SCMC studies
- Effectiveness studies

4

Research project

- Research questions
- Project timeline



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Tutor Mike (Ron C. Lee • ESLfast.com)

http://www.rong-chang.com/tutor_mike.htm

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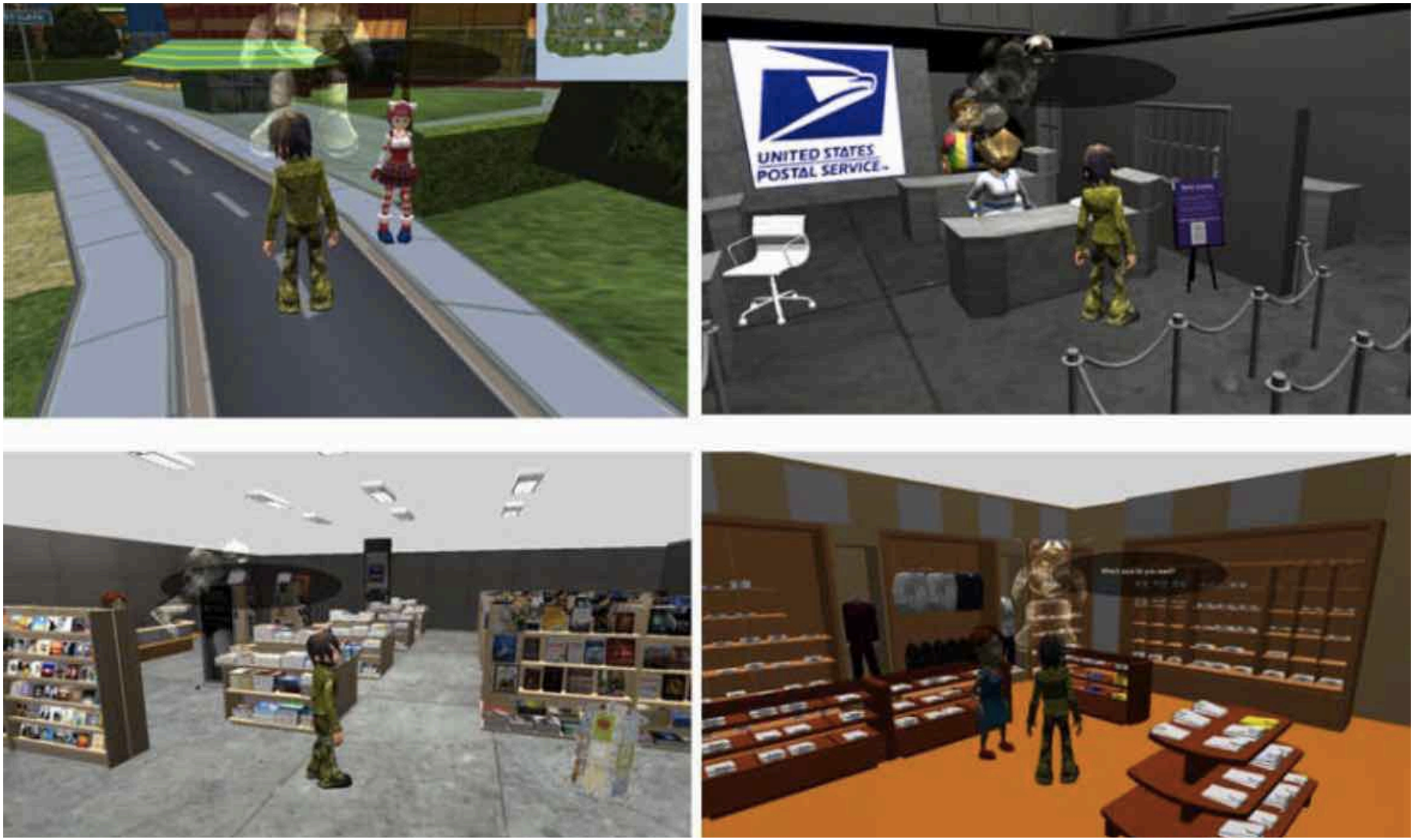
SPELL system (Morton, Gunson & Jack • University of Edinburgh)

Morton & Jack, 2005, 2010; Morton, Davidson & Jack, 2008; Morton, Gunson & Jack, 2012

POSTECH Immersive English Study (POMY)

(Lee et al. • POSTECH, South Korea)



Lee, K., Kweon, S., Lee, S., Noh, H., & Lee, G. G. (2014). POSTECH Immersive English Study (POMY): Dialog-based language learning game. *IEICE Transactions on Information and Systems*, *E97-D*(7), 1830–1841.

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POSTECH Immersive English Study (POMY)

(Lee et al. • POSTECH, South Korea)

https://sites.google.com/site/kyusonglee/

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http://dodsupport.alelo.com/tacticaloperational-language-and-culture-training

(Johnson *et al.* • ALELO)

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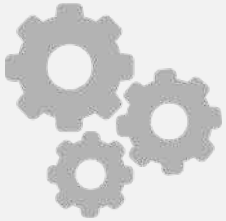

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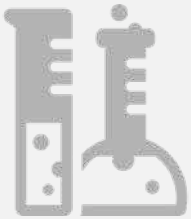

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Dialogue-based CALL: operational definition

What it is

- **dialogue**-based
  synchronous interaction between 2+ speakers, with turn-taking
- interactions **with** a computer
  computer as a **tutor**
  = **Tutorial CALL**
  (cf. Heift & Schulze, 2015)
- allowing **negotiation of outcome** through **open learner output**
  (cf. Young, 1988; Desmet & Wylin, 2015)

What it is NOT

- **item**-based
- interactions **via** a computer
  computer as a **tool** (CMC)
  = Network-Based Lang. Teaching (Kern & Warschauer, 2000)
- **predetermined outcome** through **closed learner output**

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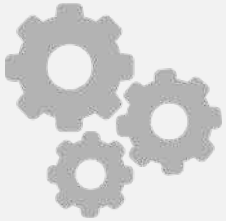

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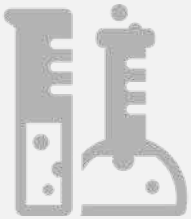

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Towards a typology of DBCALL systems

		Fixed script	Reactive systems	Graph-based systems	Plan-based systems
	Systems	•Prescripted dialogues (Saybot…)	• Question-answering (Siri…) • Chatbots (ALICE, Tutor Mike)	Most (spoken) dialogue systems POMY (Lee et al. 2014) SPELL (Morton & Jack)	• DEAL? (Wik, 2011) • [X] (Zacharski, 2002) • … ?
Instruction design	**Type of activity**	**Exercise**	**Exercise**	**Activity**	**Task**
	Context	Fixed sequence	Isolated pairs	Abstract situation	Situation + agents' objectives
	Negotiation of outcome	None	Low	Medium	High
System design	**Dialogue modeling**	**Script**	**Adjacency pairs**	**Finite-state**	**Dialogue state and planning**
	Initiative	System	User	System (±)	Mixed (±)
	Conversational memory	Fixed script	None (amnesia)	State position in the graph	Complex representation of dialogue state

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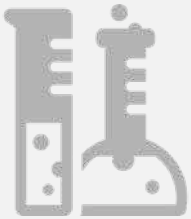

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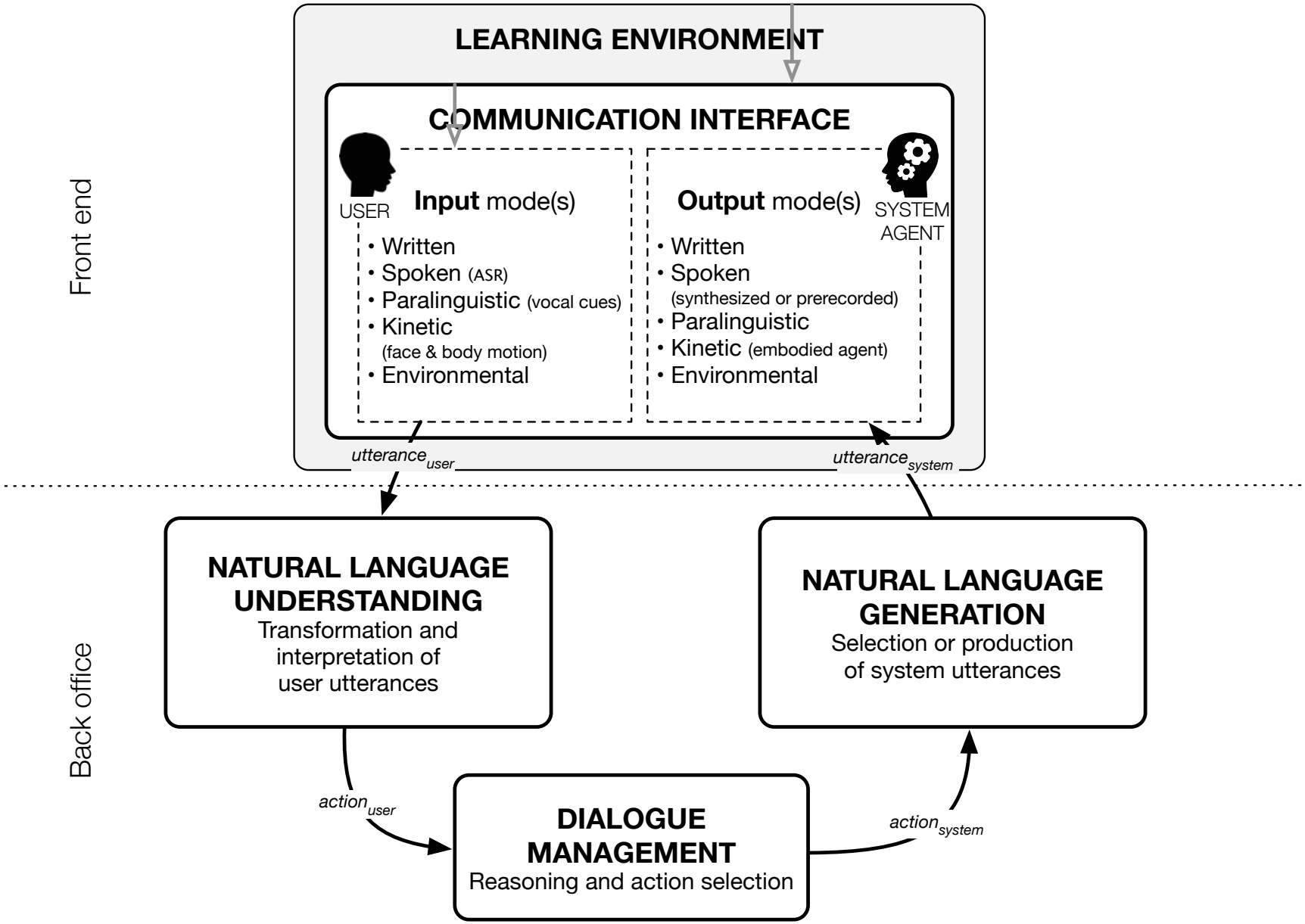
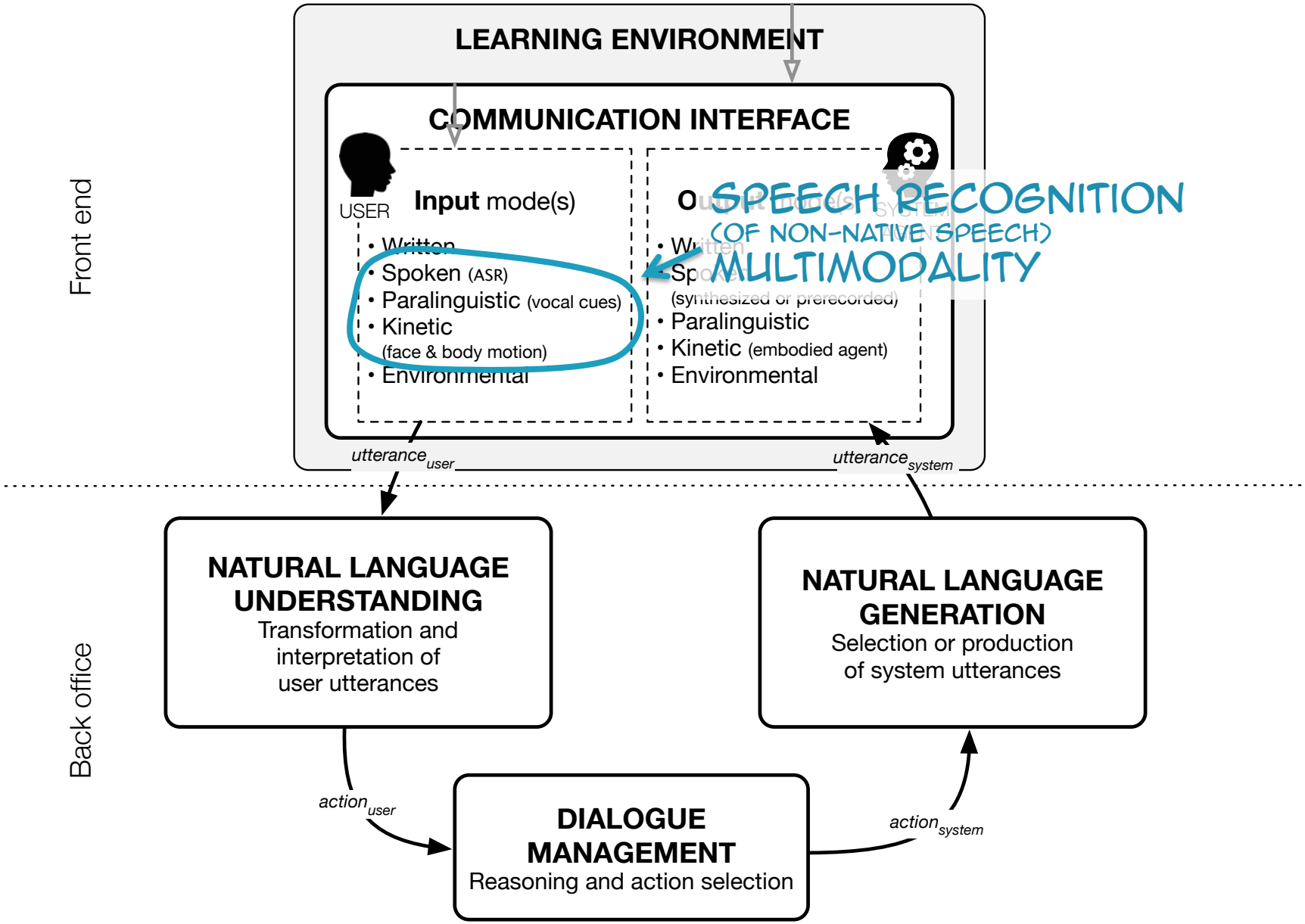
Generic NLP architecture



Technological challenges



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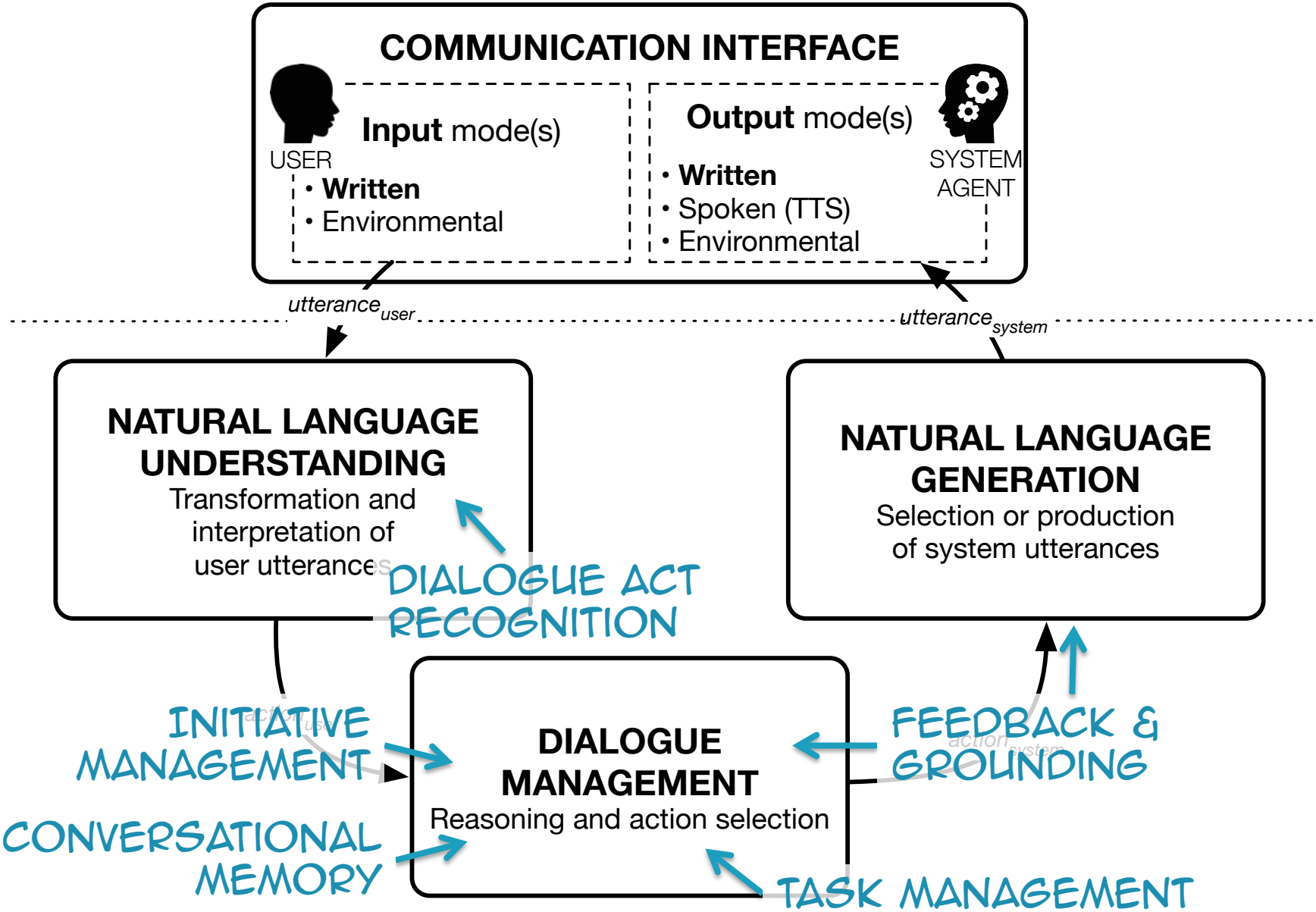

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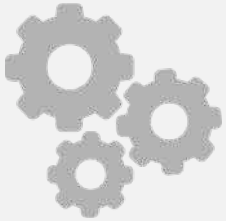

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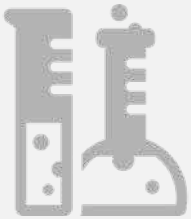

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Second language acquisition theories

- Output Hypothesis (Swain, 1985, 2005; Mackey et al., 2000)
  - noticing
  - hypothesis testing
- Interaction Hypothesis (Long, 1996; Mackey & Goo, 2007)
  - negotiation of meaning
- Cognitive-interactionist perspective (Levelt, 1989; Ellis, 1994)

SCMC (chat) ⇒ L2 development

Synchronous Computer-Mediated Communication

- SCMC: ↗ L2 oral proficiency
  (Payne & Whitney, 2002; Payne & Ross, 2005;… Ziegler, 2013; Lin, 2015)

- SCMC: ↗ L2 oral proficiency, more than face-to-face
  (Sykes, 2005; Lin, 2015; although not confirmed by Ziegler, 2013)

- Why?
  - **attention** to form (**noticing** and **feedback**)
  - lower anxiety levels (Satar & Özdener, 2008)

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DBCALL ⇢ L2 development

- In **foreign** language teaching contexts: **interactions** in L2 often **very rare**

Dialogue-based CALL allows for

- **Autonomous practice** in a communicative, cultural, authentic and interactive task
  (Wachowicz & Scott, 1999; Fryer & Carpenter, 2006)

DBCALL: advantages over SCMC

- Low anxiety environment
  (↗ willingness to communicate)
- Available at any moment
- Learner can go at his own pace (repeat, etc.)
- Potentially fully controllable learning environment
  (e.g. feedback, learner modeling and adaptivity, motivational support)

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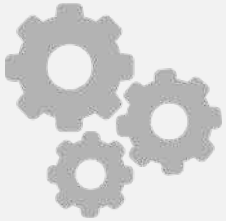

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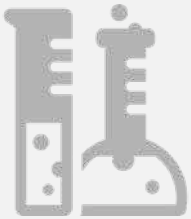

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Systematical literature review: Papers collection

- Systematical search on scientific databases:
  - Web of Science, Scopus and ProQuest
  - keywords: `chat(ter)bot, conversational agent/companion/system, dialogue system, pedagogical/dialog/virtual agent, dialogue game, human-computer dialogue (…) +`
    `language learning/acquisition/teaching, second/foreign language, ESL|EFL, English teaching/learning (…)`
  - [results: 604 / 494 / 1003 hits]
- Ancestry (citing) and forward (cited in) search
- Filtering through inclusion & exclusion criteria
- Final selection:
  139 published peer-reviewed papers
  from 1982 to (April) 2015

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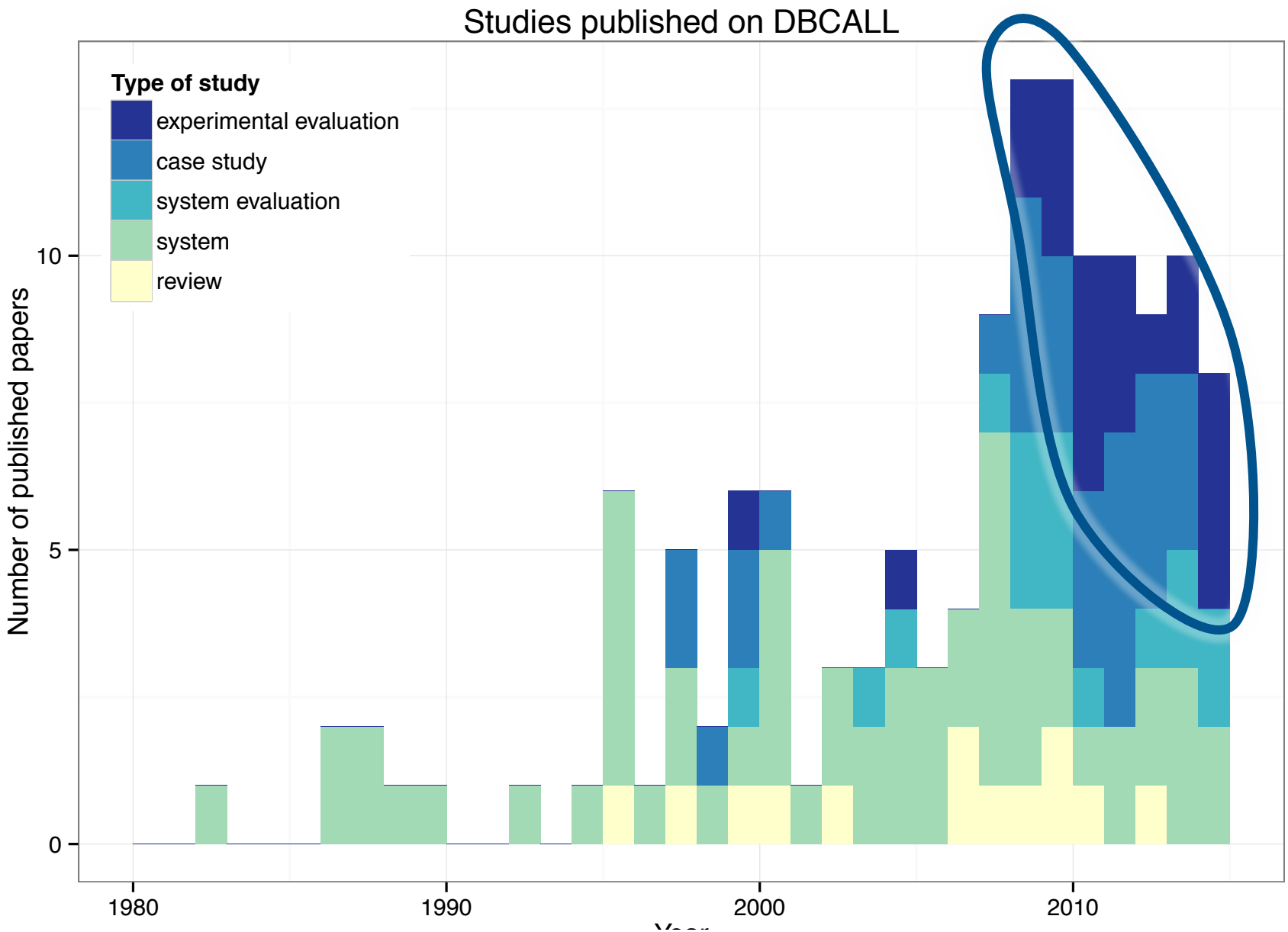
Papers collection



Effectiveness studies on DBCALL

Number of studies by dependent variables and methodological design	**Perceived usefulness**	**Impact on motivation and attitude**	**Impact on language knowledge and skills**
Post	14	1	3
Pre-post		7	22
Pre-post + control		4	12
Pre-post + control with significant results		0	2

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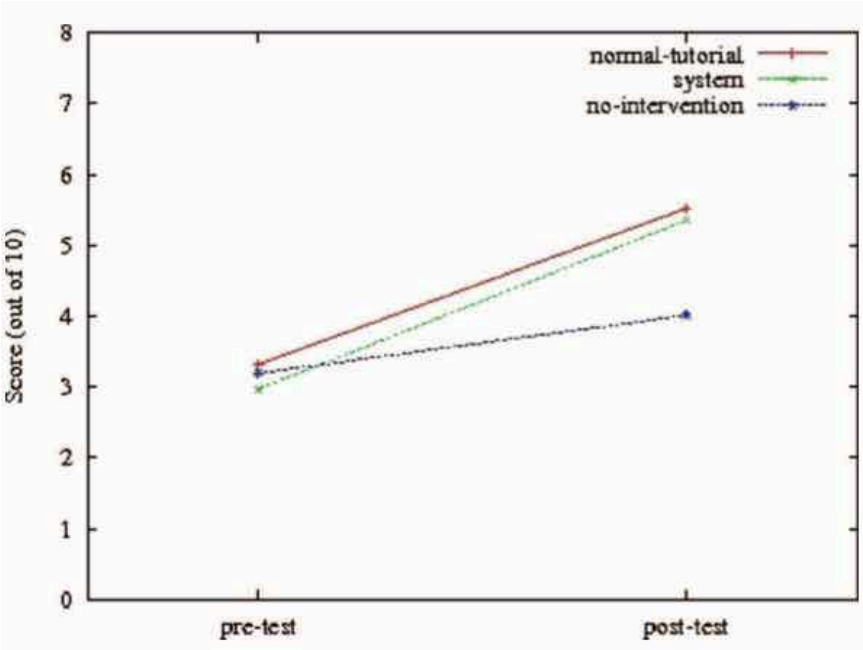
Impact on learning performance: studies

- Vlugter, Knotta, McDonald & Hall (2009)
  [Te Kaitito]: written DBCALL in Māori
  - compares DBCALL tutorial with in-class tutorial about Māori personal pronoun system



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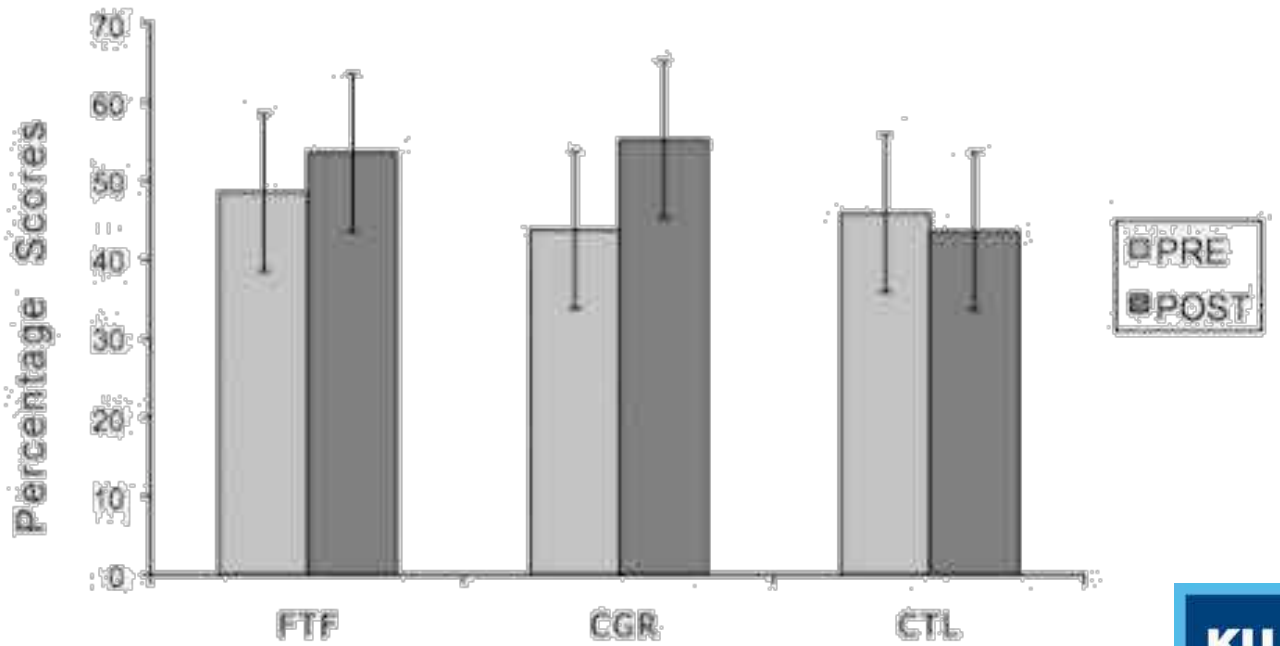
Impact on learning performance: studies

- Petersen (2010)

  [Sasha]: written DBCALL system in English providing corrective feedback (recast) and answering questions in a "spot the differences" task

  - compares Sasha vs. face-to-face interaction
    => impact on question construction



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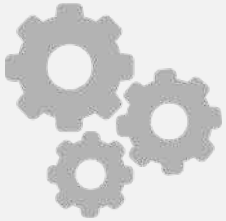

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Research questions

- What is the **best suitable natural language processing approach** to design a task-oriented dialogue system framework for language learning?
- What is the **internal efficiency** of such a task-oriented dialogue system, measured in terms of accuracy, meaningfulness, conversational coherence and ability to lead the conversation to its final purpose?
- What is the **effectiveness** of this task-oriented dialogue system for improving the **L2 proficiency** of the users, measured in terms of willingness to communicate, fluency and ability to reuse specific structures?
  How does it compare with
  a) an open-ended, free topic, conversation with an existing chatbot;
  b) a similar task-oriented conversation through a branching dialogue with previously written utterances selection;
  c) a similar task-oriented chat with a native speaker?

Project timeline

2014-2015

Literature review

- Systematical literature review
- Qualitative interpretation, supported by exploratory quantitative methods (PCA, clusters, co-citation…)
- Research synthesis publication

2015-2016

Tasks design & system programming

Tasks design

System development

- OpenDial framework
- Probabilistic rules
- Handcrafted rules + Wizard-of-Oz corpus

System evaluation

- Piloting
- PARADISE evaluation
- Coherence
- Ability to achieve task

2016-2018

Experimental evaluation

Pre-post-control experiment

Measurements:

- Willingness to communicate
- Ability to reuse spec. struct.
- Fluency in oral interview

Treatments:

- EG1: proper dialogue system
- EG2: chatbot
- EG3: branching dialogue
- CG: chat with native speaker

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<User> Thank you for your attention!
<Mike> You are very welcome.
<User> Any questions?
<Mike> Not any more.
<User> Mike, let them speak.
<Mike> What is one luxury item you really want to have?
<User> Mike, that's enough.
<Mike> I agree.

Serge Bibauw

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